Laplace Domain

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Laplace Domain

- Laplace transform
- Transfer functions
- Block diagrams
- Closed-loop systems (feedback)
- Frequency response
- Bode plots
- Bandwidth
- Elements of control loops

Why the Laplace Transform?

- Converts differential equations (dynamic systems) into algebraic equations → easier to solve
- Simplifies analysis and design
- Natural language of control theory

Reminder: Fourier Transform

Transformation from time to frequency domain:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

Tells which frequencies ω are present in a signal or system

To be exact, tabulation of both magnitude and phase response at each frequency → complex numbers:

$$F(\omega) = |F(\omega)|e^{j\phi}$$

- $|F(\omega)|$: how much of that frequency in the signal (how much the system scales each frequency)
- ϕ: time shift of that frequency (how much the system shifts each frequency)

Idea of Laplace Transform

Extend Fourier transform to include information about growth/decay behavior

→ Add exponential weighting:

$$e^{-j\omega t} \rightarrow \underbrace{e^{-j\omega t}}_{\text{oscillation}} \cdot \underbrace{e^{-\sigma t}}_{\text{growth/decay}}$$

Combine frequency ω and growth/decay rate σ:

$$j\omega \rightarrow s = \sigma + j\omega$$

Definition of Laplace Transform

Integral transform from time to complex frequency s:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st}\,dt$$

$$s = \sigma + j\omega$$

Back transformation:

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j}\int_{\sigma-\infty}^{\sigma+\infty} F(s)e^{st}\,ds$$

Example

Laplace transform of t:

$$\mathcal{L}\{t\} = \int_0^\infty t \cdot e^{-st}\, dt$$

Using partial integration: $\int u\, dv = uv - \int v\ du$

$$u = t \quad \Rightarrow \quad du = dt$$
$$dv = e^{-st}\, dt \quad \Rightarrow \quad v = -\frac{1}{s} e^{-st}$$

$$\Rightarrow \int_0^\infty t \cdot e^{-st}\, dt = \underbrace{\left[-\frac{t}{s} e^{-st}\right]_0^\infty}_{=0} + \int_0^\infty \frac{1}{s} e^{-st} dt = -\frac{1}{s^2} [e^{-st}]_0^\infty = \frac{1}{s^2}$$

Laplace of Common Signals

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Function	Time domain	Laplace domain
Unit Step	$u(t)$	$\frac{1}{s}$
Delayed unit step	$u(t-\tau)$	$\frac{1}{s}e^{-\tau s}$
Unit impulse	$\delta(t)$	1
Delayed unit impulse	$\delta(t-\tau)$	$e^{-\tau s}$
Ramp	$t \cdot u(t)$	$\frac{1}{s^2}$
Exponential decay	$e^{-\alpha t} \cdot u(t)$	$\frac{1}{s+\alpha}$
Sine	$\sin \omega t \cdot u(t)$	$\frac{\omega}{s^2+\omega^2}$
Cosine	$\cos \omega t \cdot u(t)$	$\frac{s}{s^2+\omega^2}$

Properties of Laplace Transform

Linearity: $\mathcal{L}\{a \cdot f_1(t) + b \cdot f_2(t)\} = a \cdot F_1(s) + b \cdot F_2(s)$

Convolution: $\mathcal{L}\{f_1(t) * f_2(t)\} = \mathcal{L}\left\{\int_0^t f_1(\tau) \cdot f_2(t-\tau)d\tau\right\} = F_1(s) \cdot F_2(s)$

Derivative: $\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0)$

Integration: $\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \mathcal{L}\{\,(u * f)(t)\} = \frac{1}{s}F(s)$

Frequency Shifting: $\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$

Time Shifting: $\mathcal{L}\{f(t-a) \cdot u(t-a)\} = e^{-as} \cdot F(s)$

From Differential Equation to Transfer Function

Example: PT2 system

Time domain:

$$u(t) \longrightarrow \boxed{\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2 y(t) = \omega_n^2 K u(t)} \longrightarrow y(t)$$

Frequency domain:

$$U(s) \longrightarrow \boxed{s^2Y(s) + 2\zeta\omega_n sY(s) + \omega_n^2 Y(s) = \omega_n^2 K U(s)} \longrightarrow Y(s)$$

Transfer function: $G(s) = \frac{Y(s)}{U(s)} = \frac{\omega_n^2 K}{s^2+2\zeta\omega_n s+\omega_n^2}$ (assuming zero initial conditions)

Transfer Function

Ratio of output $Y(s)$ to input $U(s)$ in Laplace domain

Fully describes input–output dynamics of system, independent of input

$$Y(s) = G(s) \cdot U(s)$$

In time domain, this corresponds to the convolution of the input with the impulse response of the system, **assuming zero initial conditions**:

$$y(t) = \int_0^t g(t - \tau) \cdot u(\tau) d\tau$$

Transfer Function for PT1 System

Differential equation:

$$\tau \cdot \dot{y}(t) + y(t) = K \cdot u(t)$$

Laplace transform:

$$\tau \cdot s \cdot Y(s) + Y(s) = K \cdot U(s)$$

Transfer Function:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1+\tau s}$$

independent of input

Example: Free Fall with Viscous Damping

$$m \cdot \dot{v}(t) + c \cdot v(t) = m \cdot g$$

where $u(t) = \begin{cases} 0, & t < 0 \\ m \cdot g, & t \geq 0 \end{cases}$ → $U(s) = m \cdot g \cdot \frac{1}{s}$

$$m \cdot s \cdot V(s) + c \cdot V(s) = m \cdot g \cdot \frac{1}{s}$$

$$V(s) = \frac{m \cdot g}{s(m \cdot s + c)} = \frac{g}{s\left(s + \frac{c}{m}\right)}$$

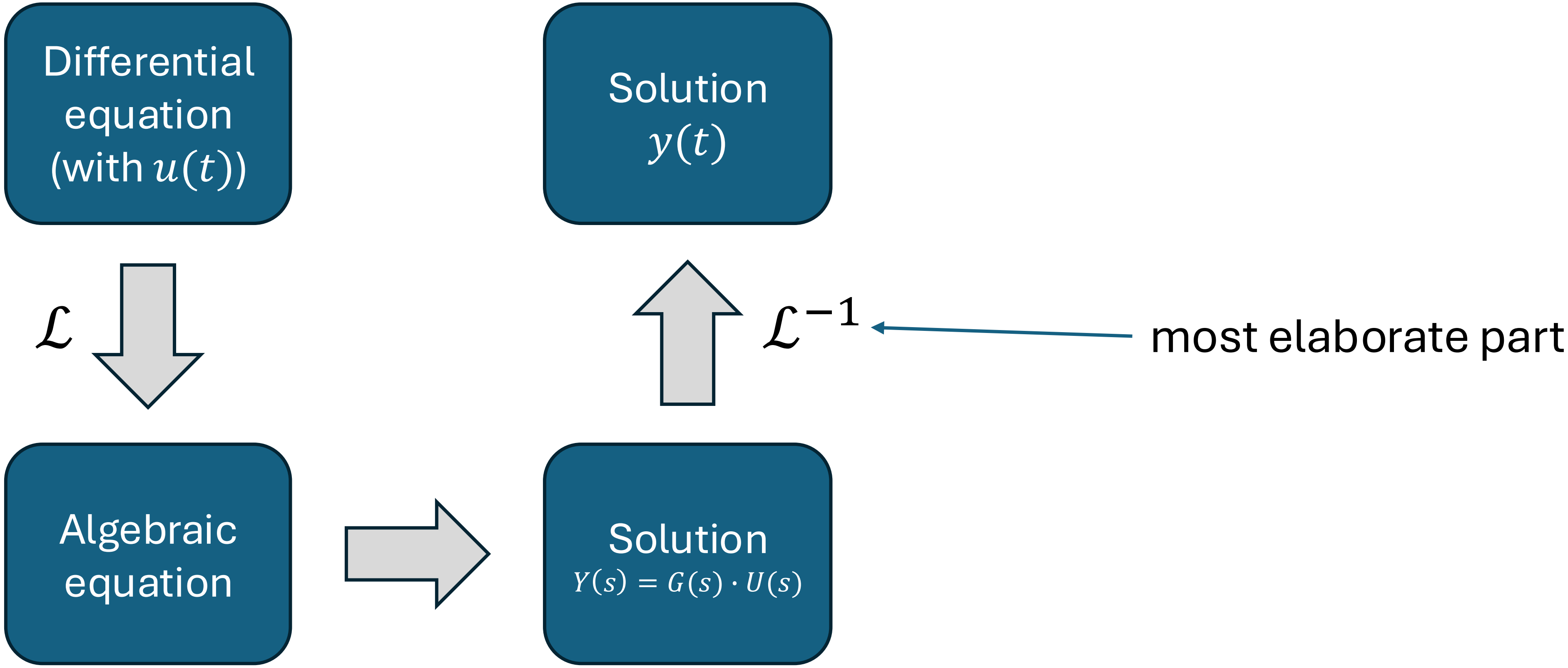
Transformation table:

$$\frac{\alpha}{s(s+\alpha)} \mapsto 1 - e^{-\alpha t}$$

→ $v(t) = g \cdot \frac{m}{c} \cdot \left(1 - e^{-\frac{c}{m}t}\right)$

$$\tau = K = \frac{m}{c}$$

System Response with Transfer Function



Transfer Function in Polynomial Form

$$G(s) = \frac{K}{1+\tau s}, \qquad G(s) = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}, \qquad \ldots$$

Input coupling:
describes how the input enters the system (can contain time derivatives)

$$\rightarrow G(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0} = \frac{b_0}{a_0}\,\frac{\frac{b_m}{b_0}s^m + \cdots + \frac{b_1}{b_0}s + 1}{\frac{a_n}{a_0}s^n + \cdots + \frac{a}{a_0}s + 1}$$

fundamental theorem of algebra

$$K = \frac{b_0}{a_0}$$

Intrinsic system dynamics:
characteristic polynomial of the system

$$\rightarrow G(s) = \frac{b_m}{a_n}\,\frac{\prod_{i=1}^{m}(s - s_{0i})}{\prod_{i=1}^{n}(s - s_{pi})}$$

roots of numerator: m zeros s_0

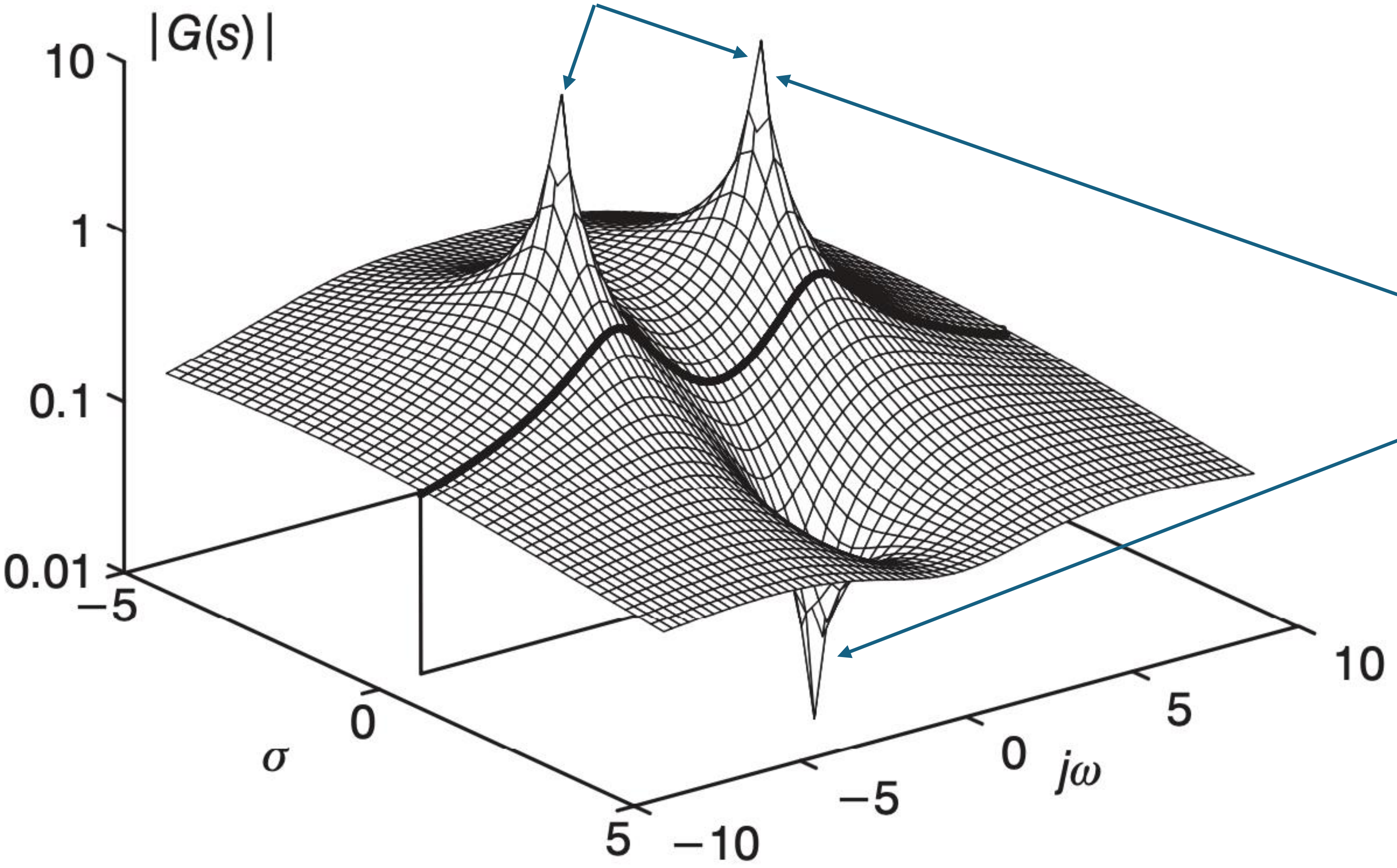
roots of denominator: n poles s_p

$$k = \frac{b_m}{a_n}$$

Causality requires $n \geq m$ (output cannot depend on derivative of input)

Graphical Interpretation of Transfer Function

conjugate pair



Contains (complex) poles and zeros

Poles and Zeros

PT1: one pole
PT2: two poles

Poles and zeros of a transfer function can be real or complex

Complex poles/zeros always appear in conjugate pairs for real physical systems (need for real coefficients in differential equations)

$$\boxed{\big(s-(\sigma+j\omega)\big)\big(s-(\sigma-j\omega)\big)=(s-\sigma)^2+\omega^2}$$

Poles determine stability and dynamics (system's natural response)

$$y_i(t)=C_i e^{s_{pi}t}, \qquad y_{\mathrm{hom}}(t)=\textstyle\sum_i y_i(t)$$

More about this later ...

Zeros shape the forced (input-driven) response

Step Response

Unit step $u(t)$ in Laplace domain: $U(s) = \frac{1}{s}$

→ $Y(s) = G(s) \cdot U(s) = \frac{1}{s} \cdot G(s)$

Example: $G(s) = \frac{1}{s+1}$

→ $Y(s) = \frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1}$ partial fraction expansion

- For A (pole at $s = 0$): multiply with s and set $s = 0$: $A = \left[\frac{1}{s+1}\right]_{s=0} = 1$
- For B (pole at $s = -1$): multiply with $s + 1$ and set $s = -1$: $A = \left[\frac{1}{s}\right]_{s=-1} = -1$

→ $Y(s) = \frac{1}{s} - \frac{1}{s+1}$ back transformation: $y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s+1}\right\} = (1 - e^{-t})$

Final Value Theorem

Steady-state value for unit step response:

$$y_\infty = \lim_{t\to\infty} y(t) = \lim_{s\to 0} s \cdot Y(s)$$

Derivation:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = \int_0^\infty \frac{df(t)}{dt} \cdot e^{-st} dt = sF(s) - f(0)$$

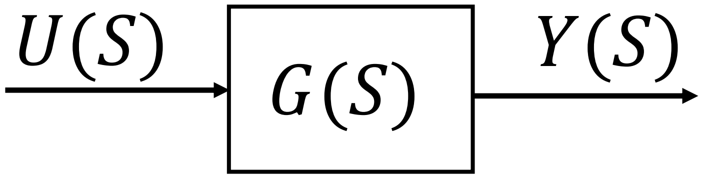
$$\Rightarrow \lim_{s\to 0} \int_0^\infty \frac{df(t)}{dt} \cdot e^{-st} dt = \lim_{s\to 0}[sF(s) - f(0)\,]$$

using: $\lim_{s\to 0} \int_0^\infty \frac{df(t)}{dt} \cdot e^{-st} dt = [f(t)]_0^\infty = \lim_{t\to\infty} f(t) - f(0)$

$$\Rightarrow \lim_{t\to\infty} f(t) - f(0) = \lim_{s\to 0}[sF(s) - f(0)\,]$$

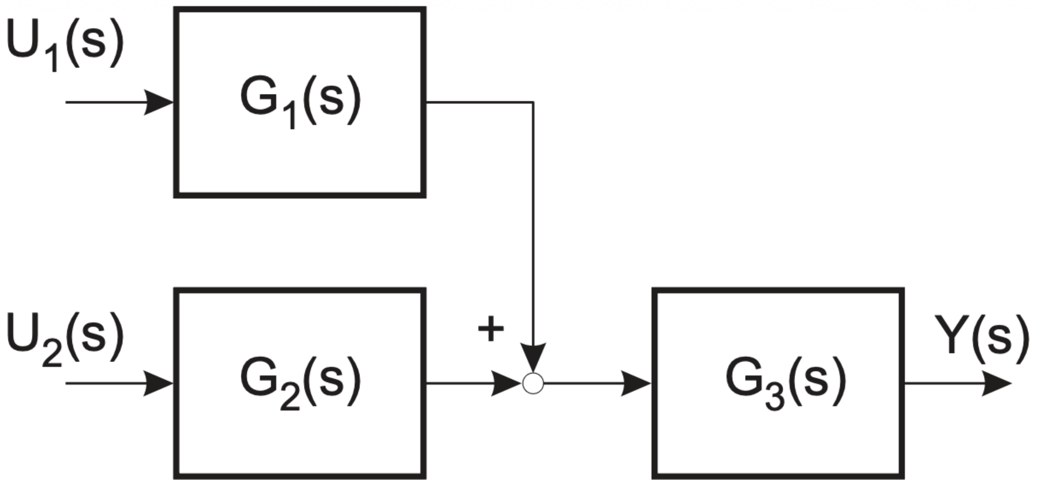
$$\Rightarrow \lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

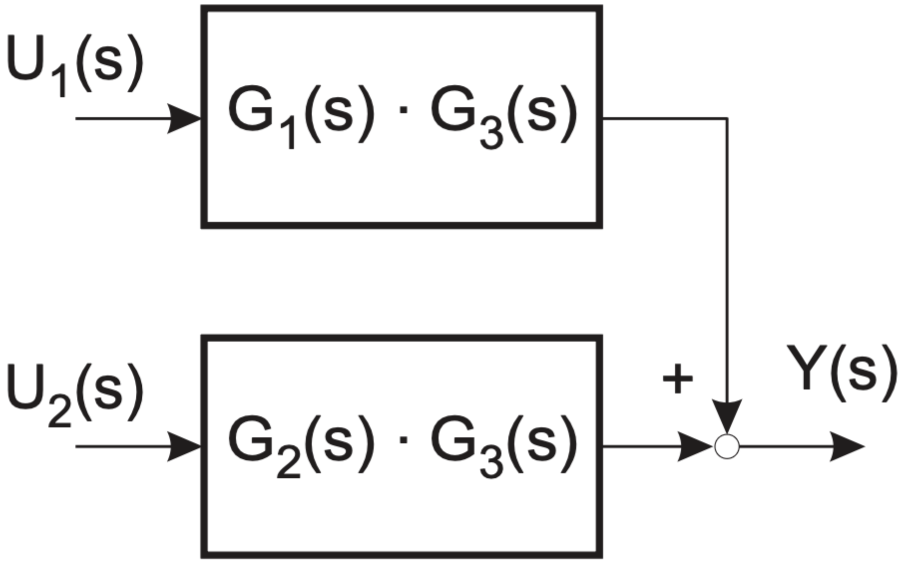
Block Diagrams

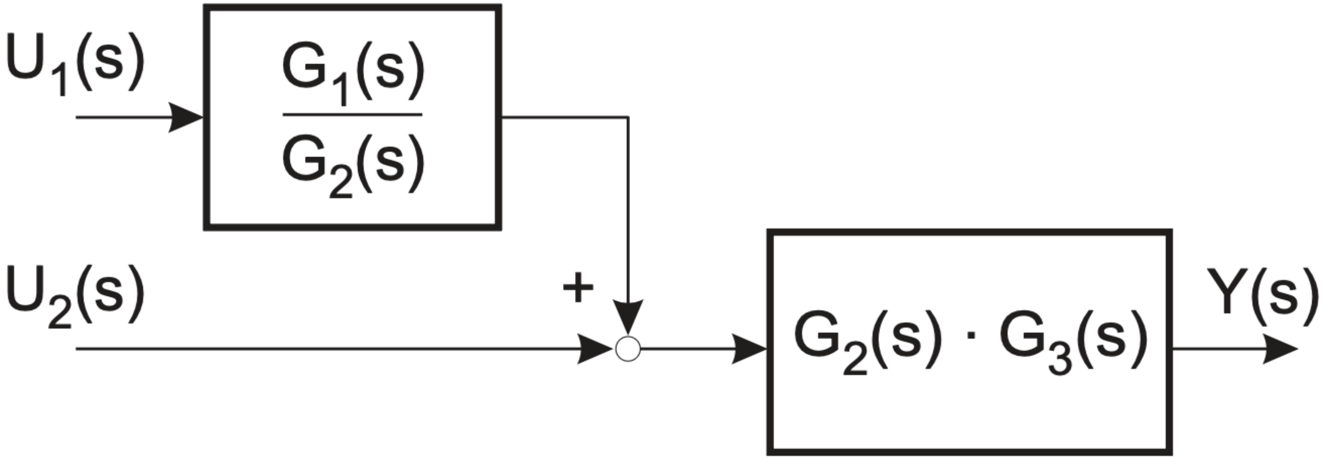


Modular representation

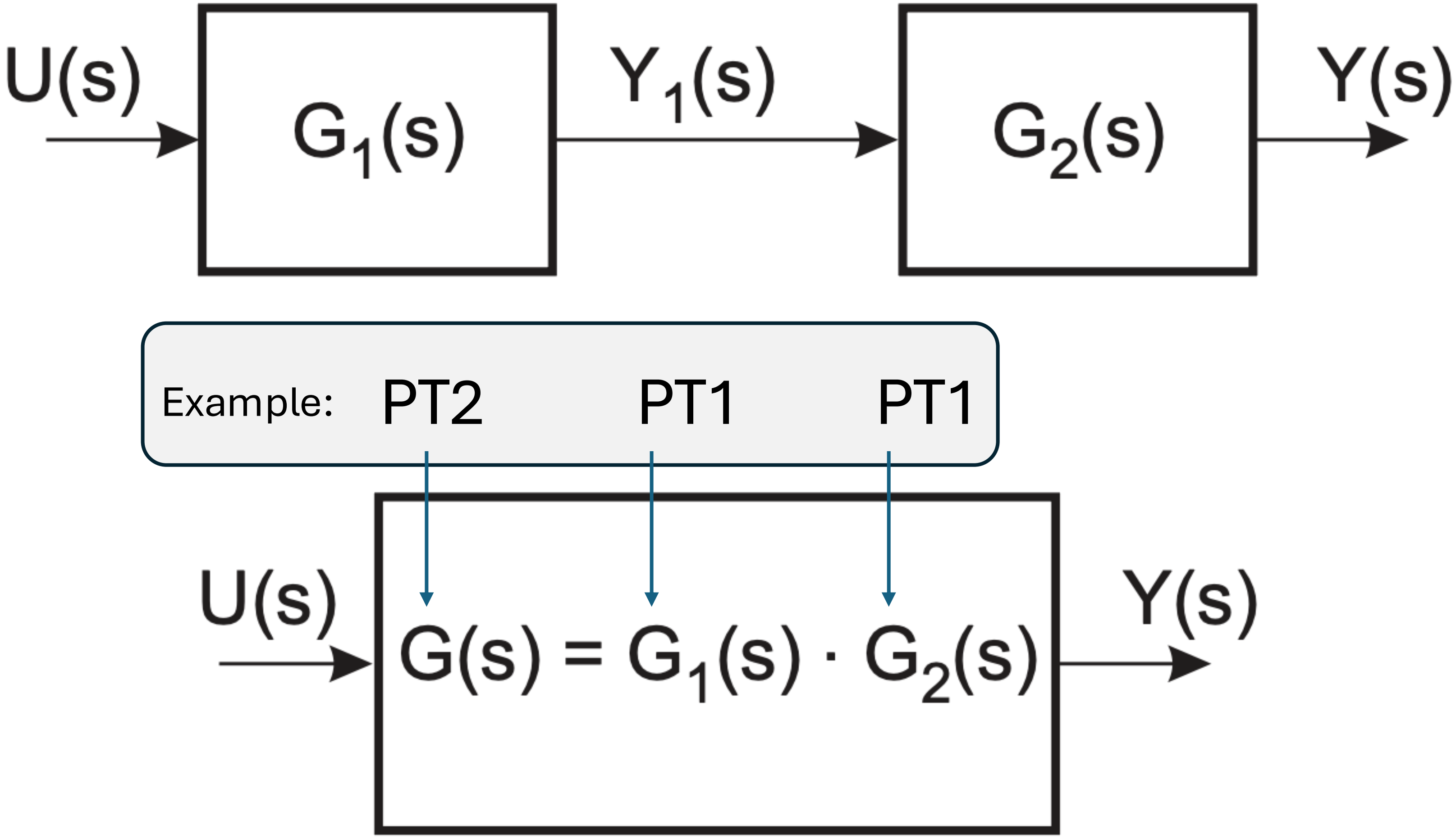
Block Diagram Algebra



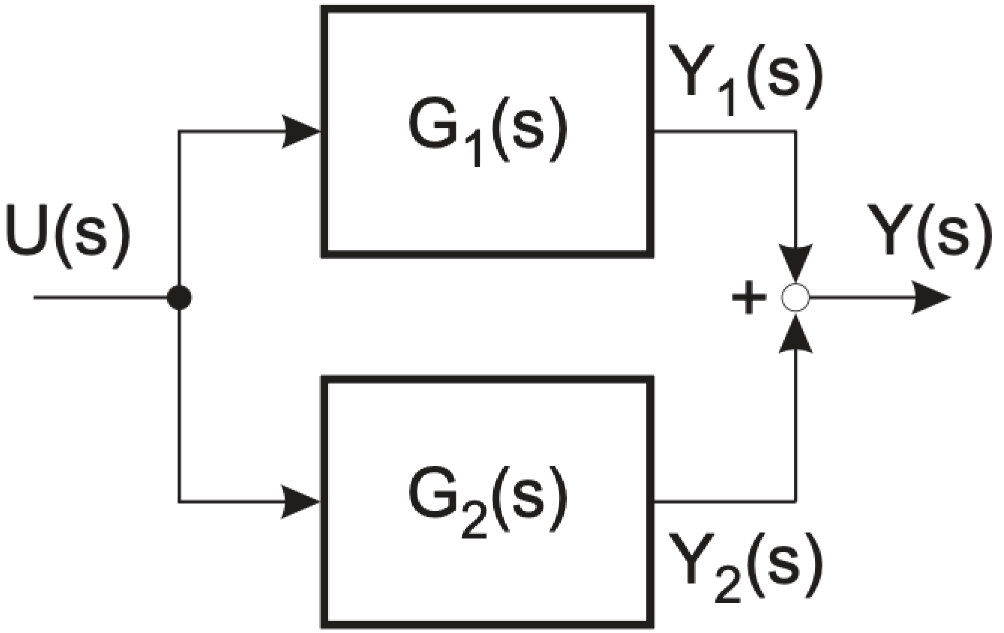


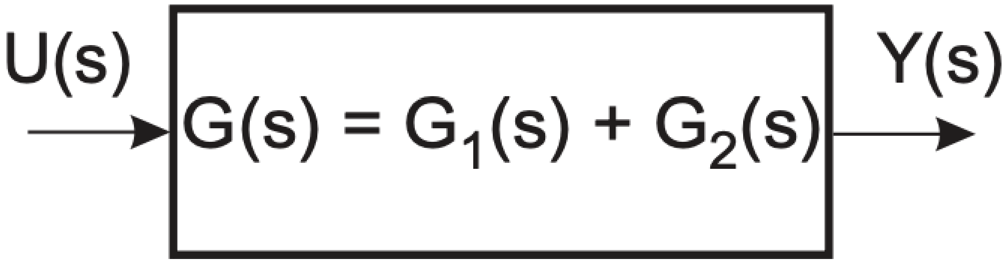


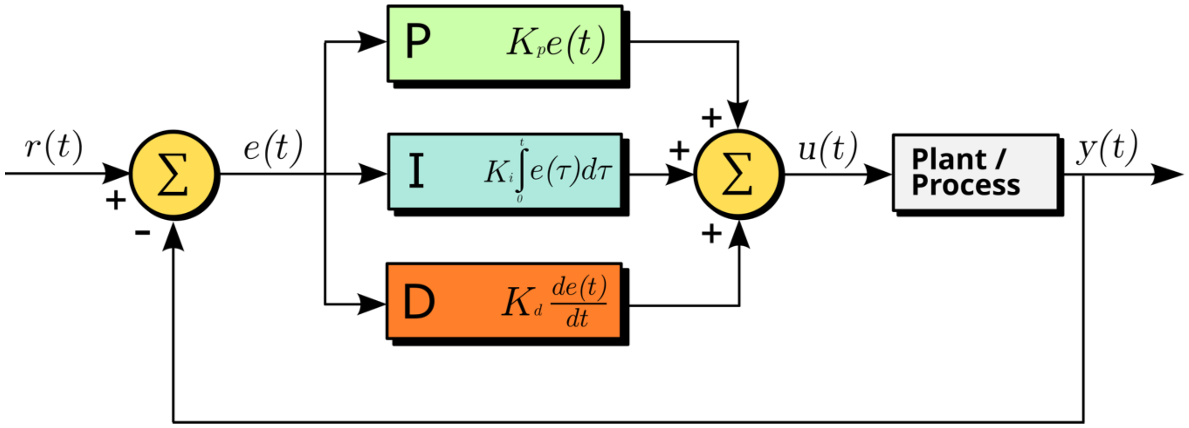
Series Connection



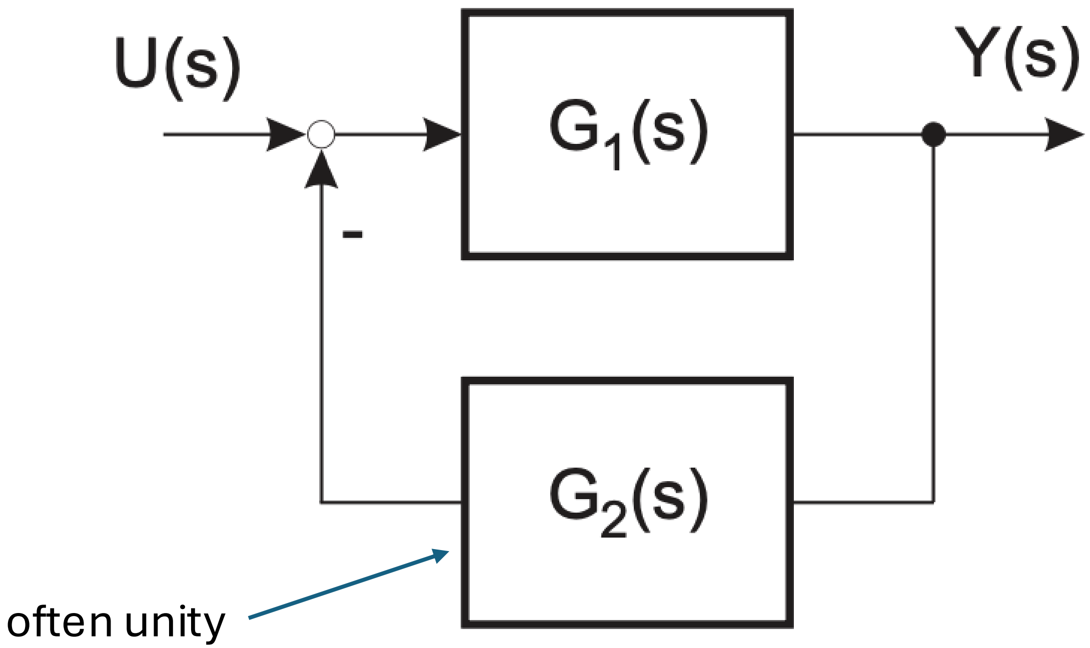
Parallel Connection

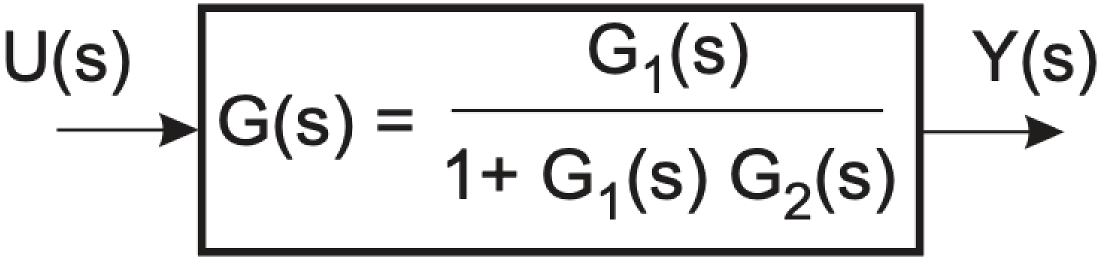






Feedback Connection





$$Y(s) = G_1(s) \cdot E(s)$$
$$E(s) = U(s) - G_2(s) \cdot Y(s)$$

$$\Rightarrow Y(s) = G_1(s) \cdot \big(U(s) - G_2(s) \cdot Y(s)\big)$$

$$\Rightarrow Y(s) \cdot \big(1 + G_1(s) \cdot G_2\,(s)\big) = G_1(s) \cdot U(s)$$

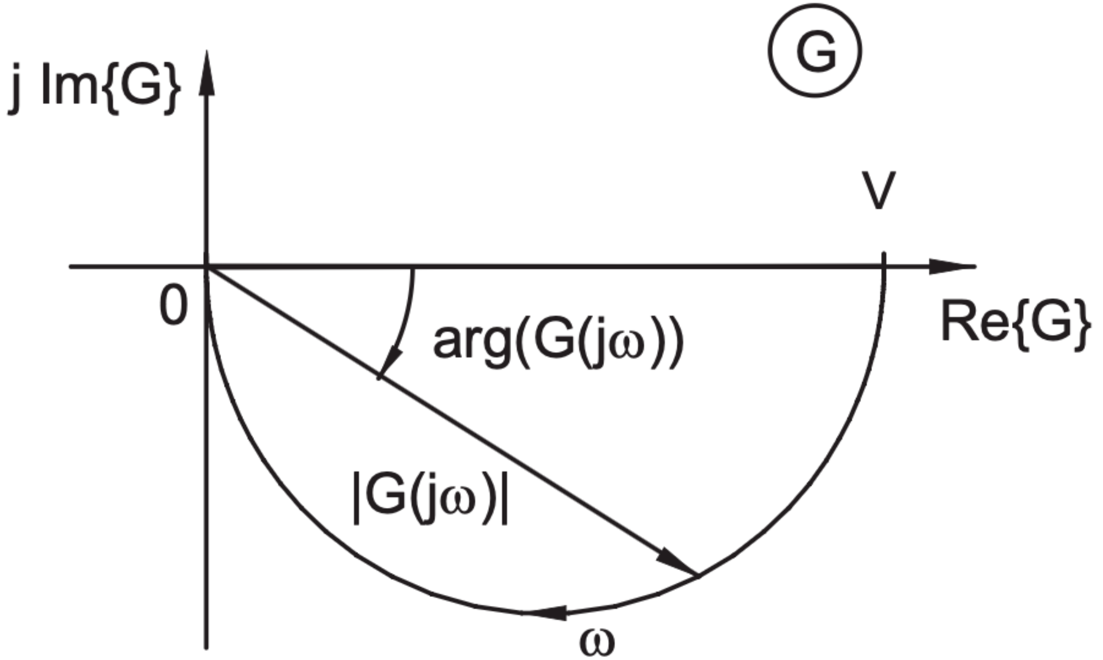
$$\Rightarrow \frac{Y(s)}{U(s)} = \frac{G_1(s)}{\big(1 + G_1(s) \cdot G_2\,(s)\big)}$$

response = transient part + steady-state part

Frequency Response

- Response to sinusoidal inputs: $G(j\omega)$
- Amplitude and phase as function of frequency ω

Nyquist diagram:



Bode Plots

Example PT1:

$G(j\omega) = \frac{K}{1+\tau j\omega}$

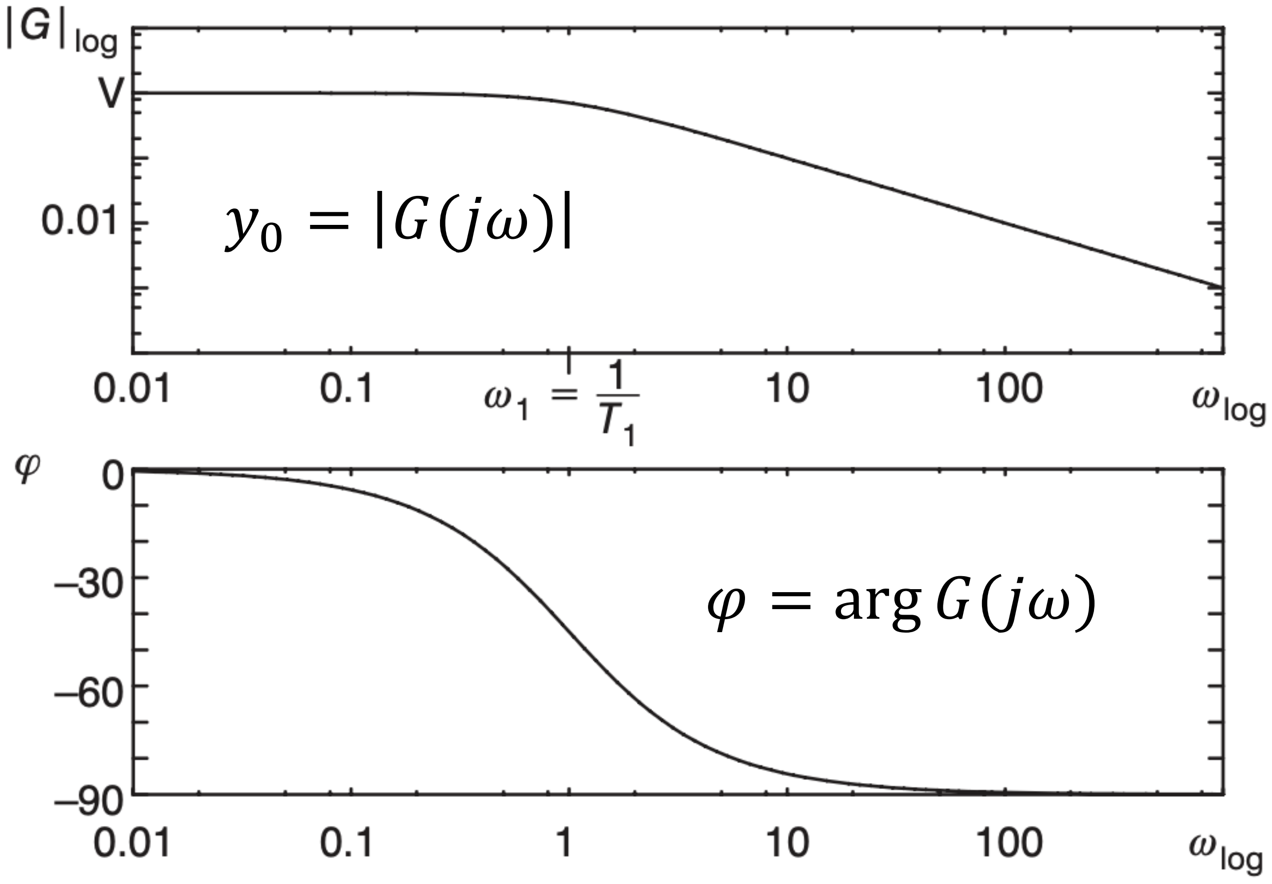
$|G(j\omega)| = \frac{K}{\sqrt{1+\tau^2\omega^2}}$

$\varphi = -\tan^{-1}\left(\frac{\omega\tau}{K}\right)$

Frequency response ($s = j\omega$):

$$u(t) = \sin \omega t$$

$$y(t) = y_0 \sin(\omega t + \varphi)$$



1. Magnitude (dB) vs frequency
2. Phase vs frequency

Both with logarithmic frequency axis

Common Magnitude Unit: Decibel

Ratio of two values of a power quantity (or one measurement P and a reference value P_0):

$$10 \cdot \log_{10}\left(\frac{P}{P_0}\right) \mathrm{dB}$$

→ A change in power by a factor of 10 corresponds to a $10\ \mathrm{dB}$

Ratio of two amplitudes ($P \sim A^2$) → root-power quantities:

$$20 \cdot \log_{10}\left(\frac{A}{A_0}\right) \mathrm{dB} \quad \text{or} \quad 20 \cdot \log_{10}(|G(j\omega)|)\ \mathrm{dB}$$

→ A change in amplitude by a factor of 10 corresponds to a $20\ \mathrm{dB}$

Bandwidth

Frequency range with good tracking (i.e., closed-loop system follows reference signal closely in shape, speed, and steady-state value)

Strong damping for higher frequencies (kind of filter on reference)

Typically consider frequency for which $|G(j\omega)|$ changes by -3 dB:

$|G(j\omega_B)| = \frac{1}{\sqrt{2}} \cdot |G(0)|$ (amplitude drops to 70.7% of its DC value)

For first-order system: $|G(j\omega)| = \frac{1}{\sqrt{1+(\omega\tau)^2}} \stackrel{\text{def}}{=} \frac{1}{\sqrt{2}} \Rightarrow \omega_B = \frac{1}{\tau}$

Indicates speed of system: higher bandwidth → faster response

But also less robust and higher noise amplification

Individual Elements of Control Loops

Polynomial form of differential equation / transfer function:

$$(a_n s^n + \cdots + a_1 s + a_0) \cdot Y(s) = (b_m s^m + \cdots + b_1 s + b_0) \cdot U(s)$$

→ Transfer functions of individual elements by setting a portion of the coefficients to zero

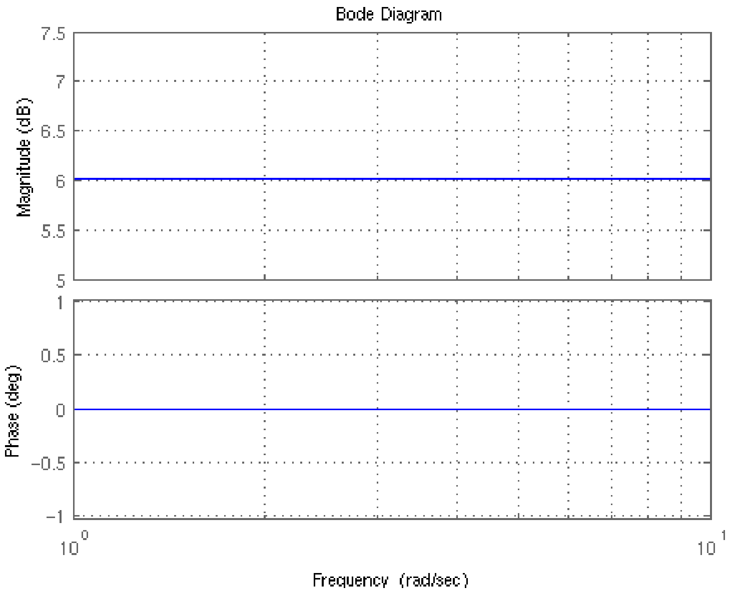
Proportional Element (P)

Consider $n = 0$:

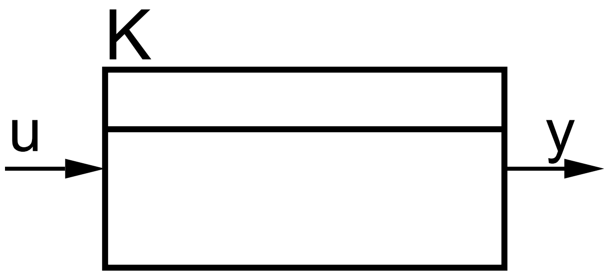
$$y(t) = \frac{b_0}{a_0} \cdot u(t) \quad \text{or} \quad Y(s) = \frac{b_0}{a_0} \cdot U(s)$$

$$\frac{b_0}{a_0} = K_P$$

(just constant gain)



Step response:



Integral Element (I)

Pole at $s = 0$

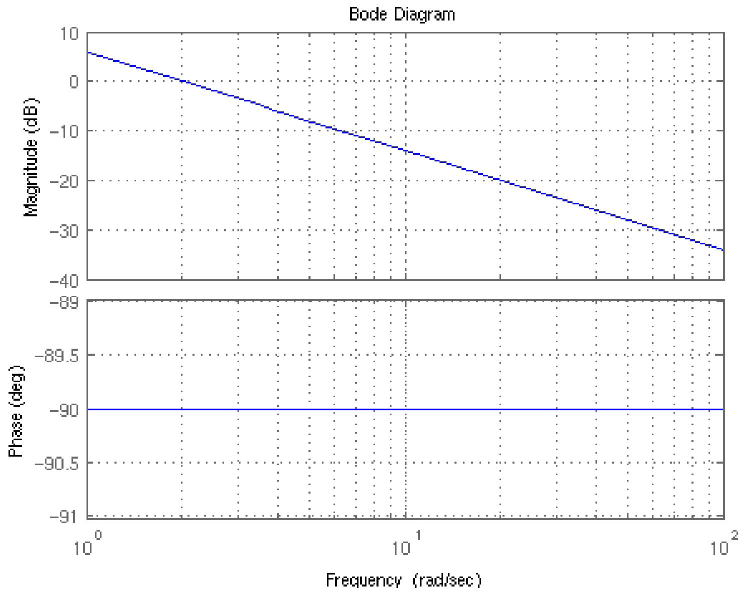
Consider $n = 1$, with $a_0 = b_1 = 0$:

$$\frac{dy(t)}{dt} = \frac{b_0}{a_1} \cdot u(t) \quad \text{or} \quad Y(s) = \frac{b_0}{a_1} \cdot \frac{1}{s} \cdot U(s)$$

$$y(t) = \frac{b_0}{a_1} \cdot \int u(t)\, dt$$

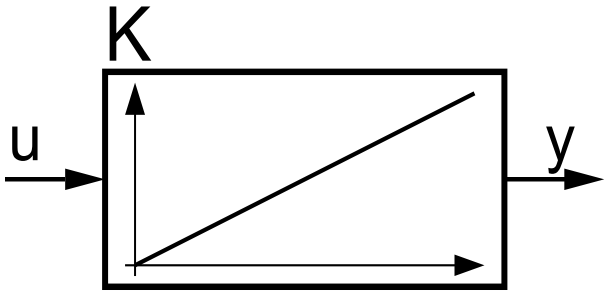
$$\frac{b_0}{a_1} = K_I = {}^1\!/_{T_I}$$

(integrator gain or integral time constant)



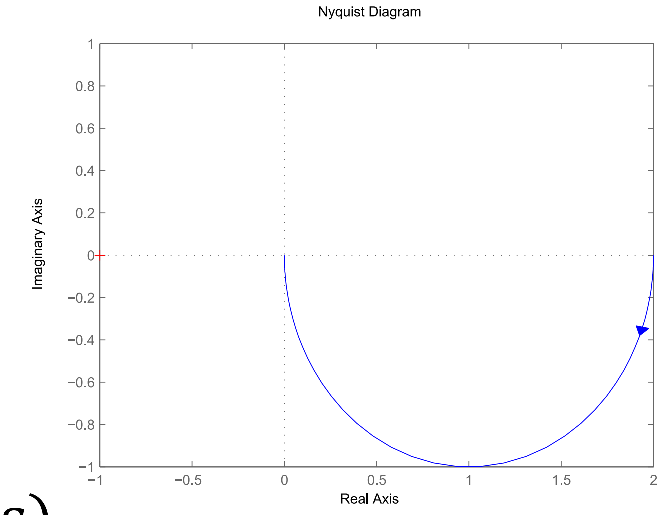
Step response:

$$\int \text{const}\, dt = t$$



Energy storage, memory

First-Order Lag Element (PT1)



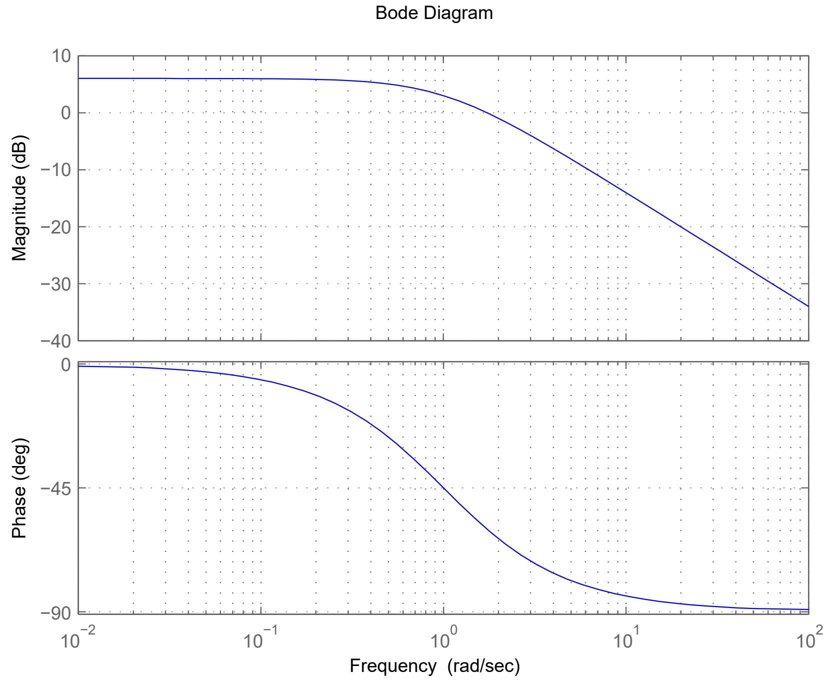
Consider $n = 1$, with $b_1 = 0$:

$$\frac{a_1}{a_0} \cdot \frac{dy(t)}{dt} + y(t) = \frac{b_0}{a_0} \cdot u(t) \quad \text{or} \quad Y(s) \cdot \left(s \cdot \frac{a_1}{a_0} + 1\right) = \frac{b_0}{a_0} \cdot U(s)$$

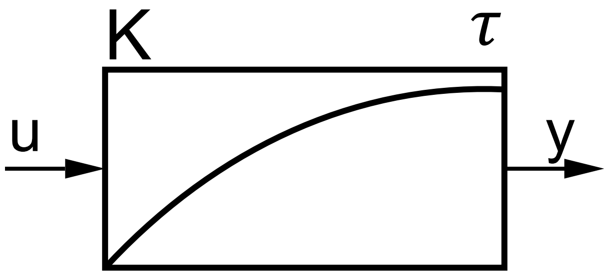
τ $\qquad$ K

$$\boxed{G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1 + \tau s}}$$

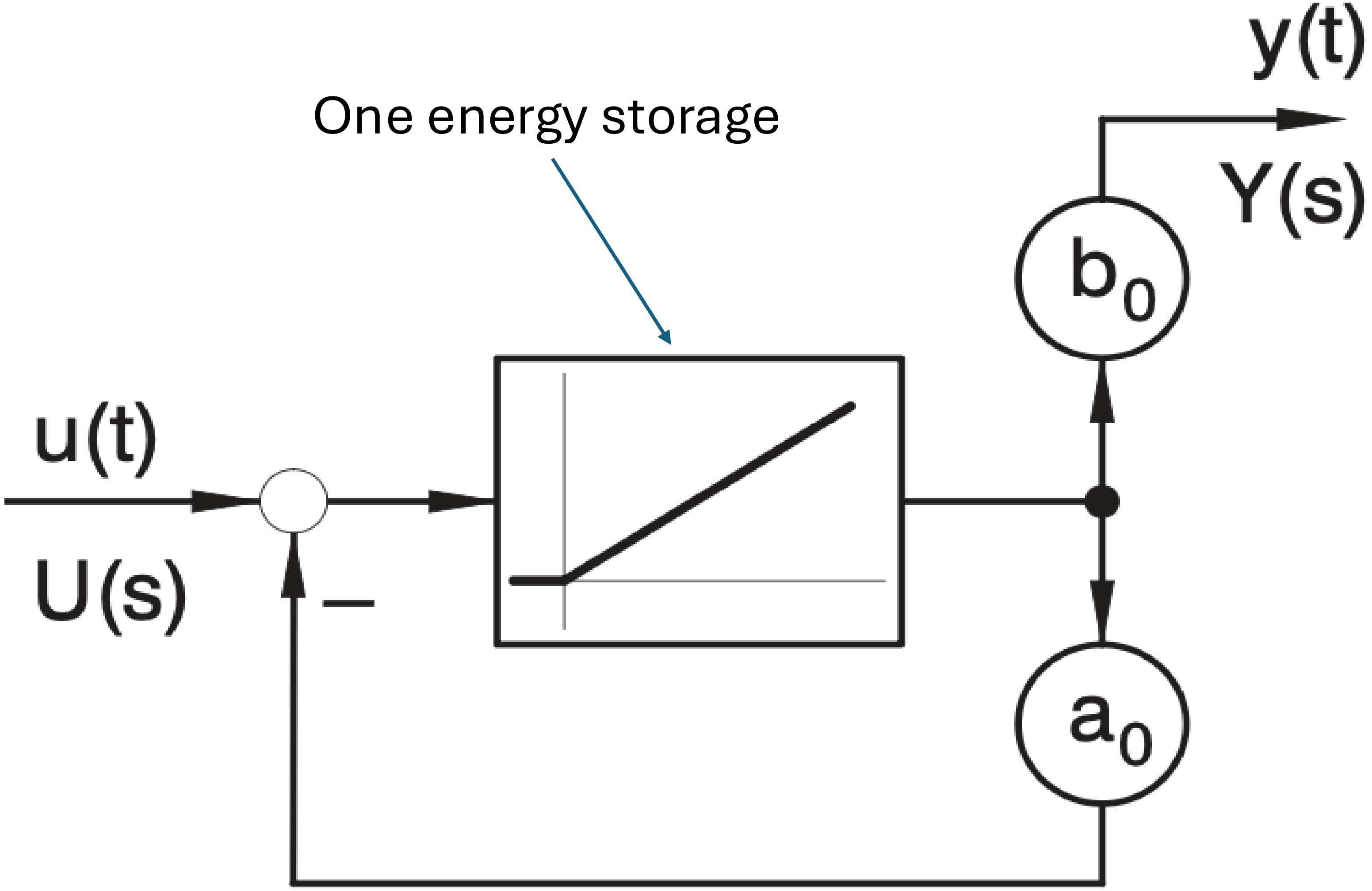
Pole at $s = -\frac{1}{\tau}$



Step response:



PT1 Signal Flow Diagram

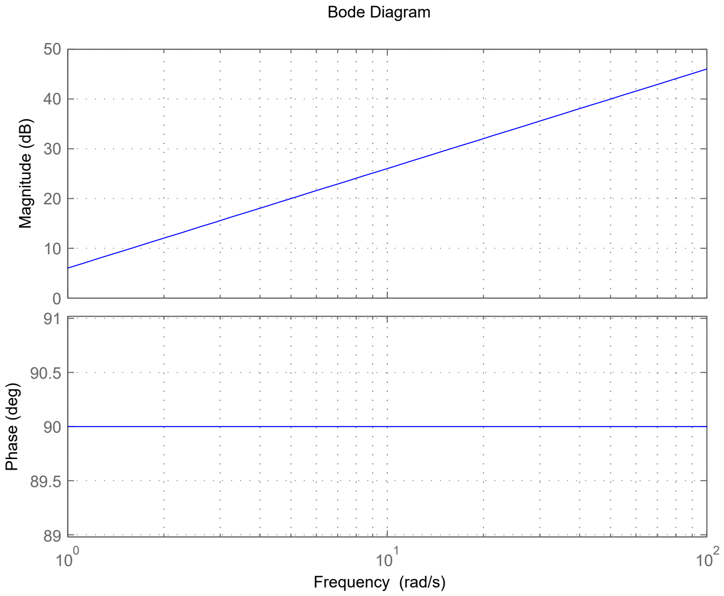


Derivative Element (D)

Amplifies high frequencies, i.e., noise
→ use filtered differentiator (DT1)

Consider $n = 1$, with $a_1 = b_0 = 0$:

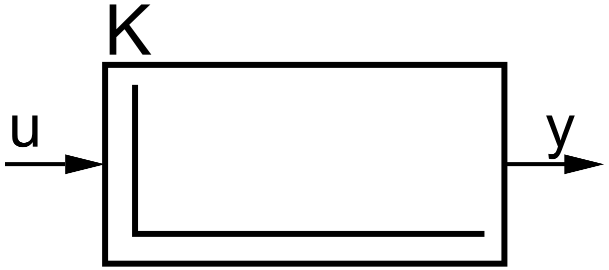
$$y(t) = \frac{b_1}{a_0} \cdot \frac{du(t)}{dt} \quad \text{or} \quad Y(s) = \frac{b_1}{a_0} \cdot s \cdot U(s)$$



$$\frac{b_1}{a_0} = K_D = T_D$$

(differentiator/derivative gain or derivative time constant)

Step response:



physically unrealizable → damping

Filtered Differentiator (DT1)

Used in PID controllers

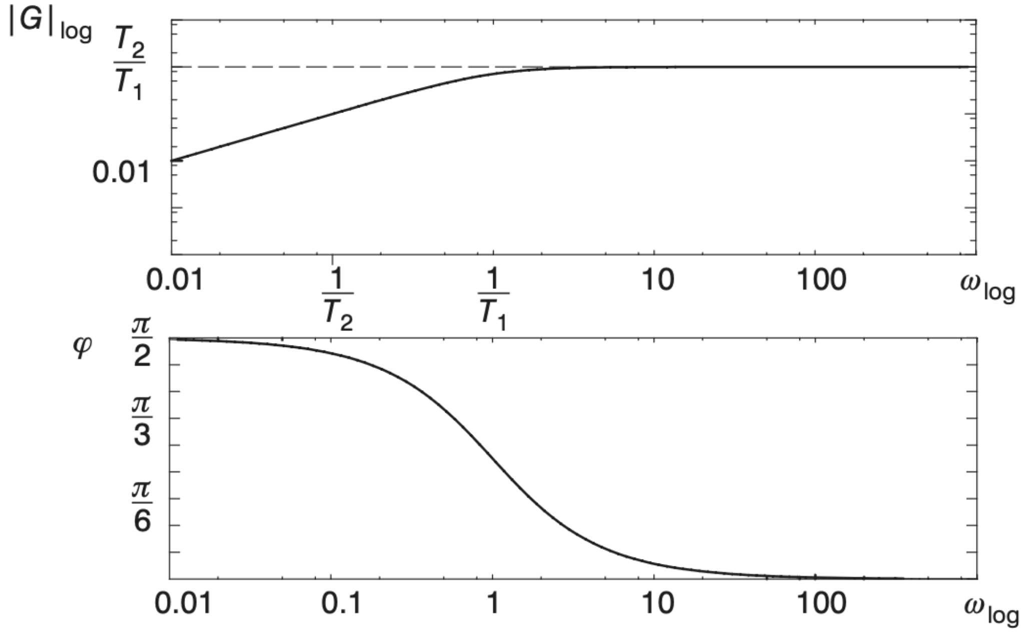
Consider $n = 1$, with $b_0 = 0$:

$$\underbrace{\frac{a_1}{a_0}}_{T_1} \cdot \frac{dy(t)}{dt} + y(t) = \underbrace{\frac{b_1}{a_0}}_{T_2} \cdot \frac{du(t)}{dt} \quad \text{or} \quad Y(s) \cdot \left(s \cdot \frac{a_1}{a_0} + 1\right) = \frac{b_1}{a_0} \cdot s \cdot U(s)$$

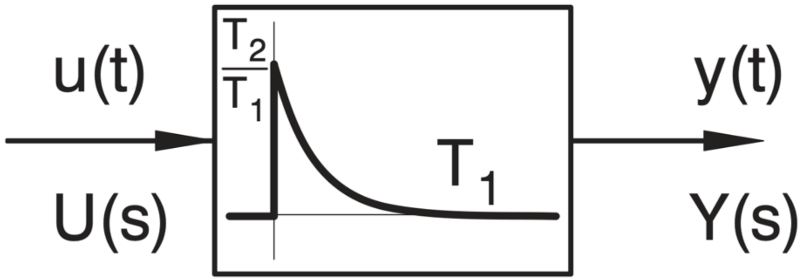
$$\boxed{G(s) = \frac{T_2 \cdot s}{T_1 \cdot s + 1}}$$

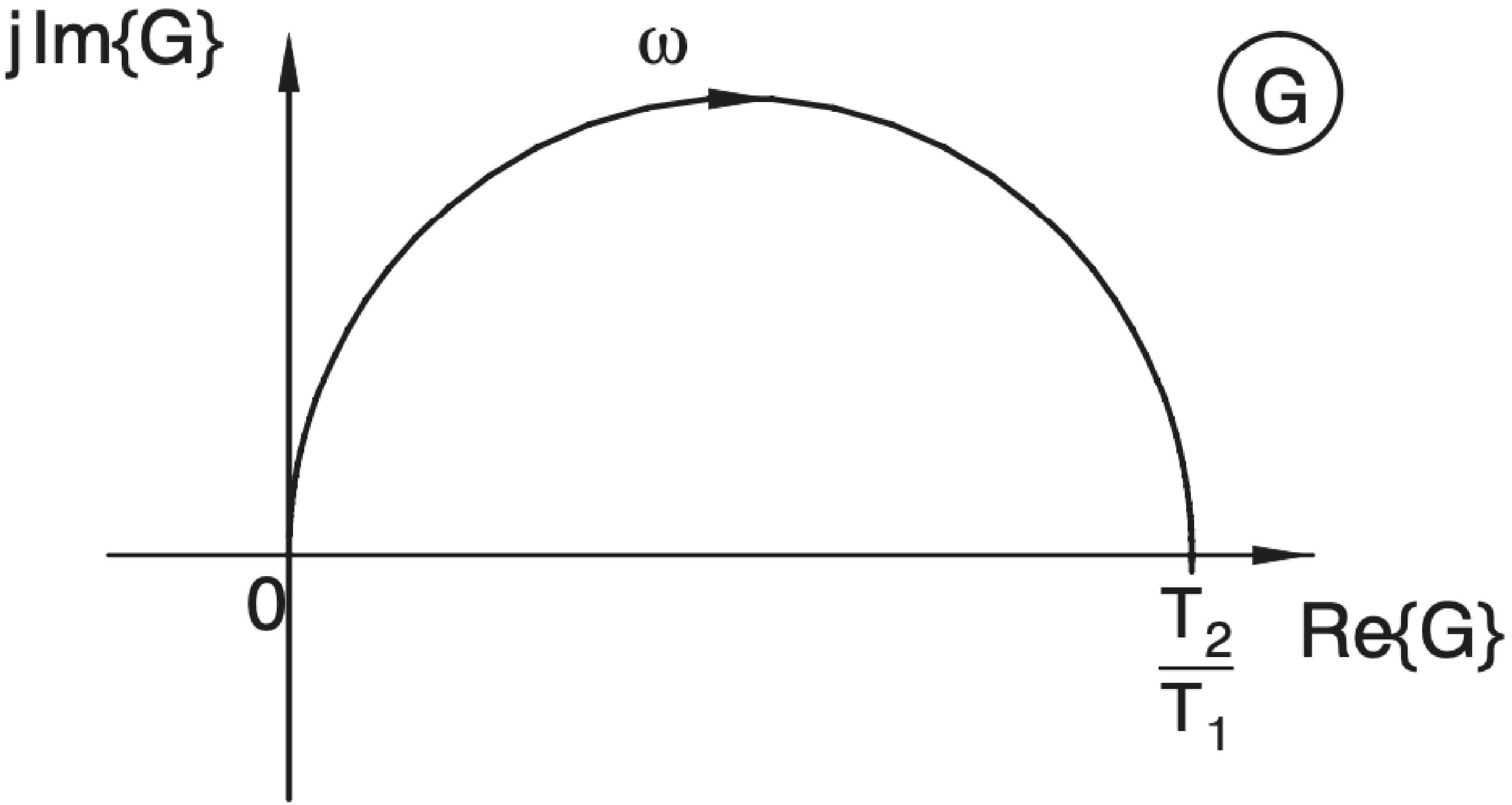
$$\lim_{T_1 \to 0} G(s) = T_2 \cdot s$$

→ ideal D element

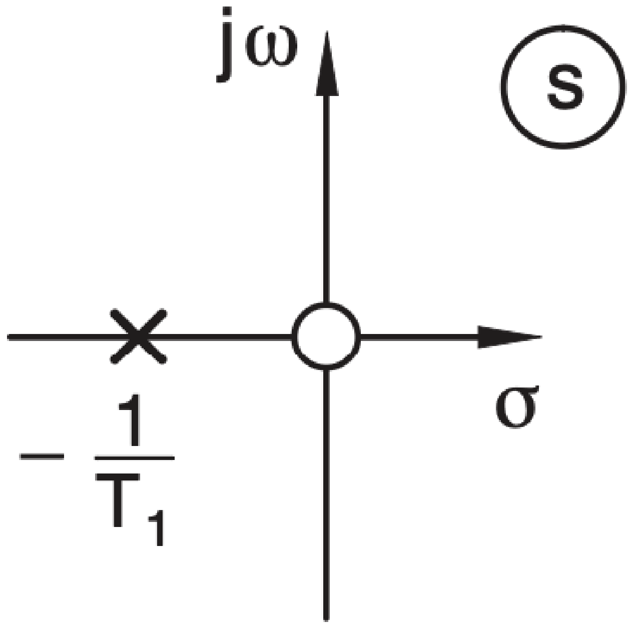


Step response:



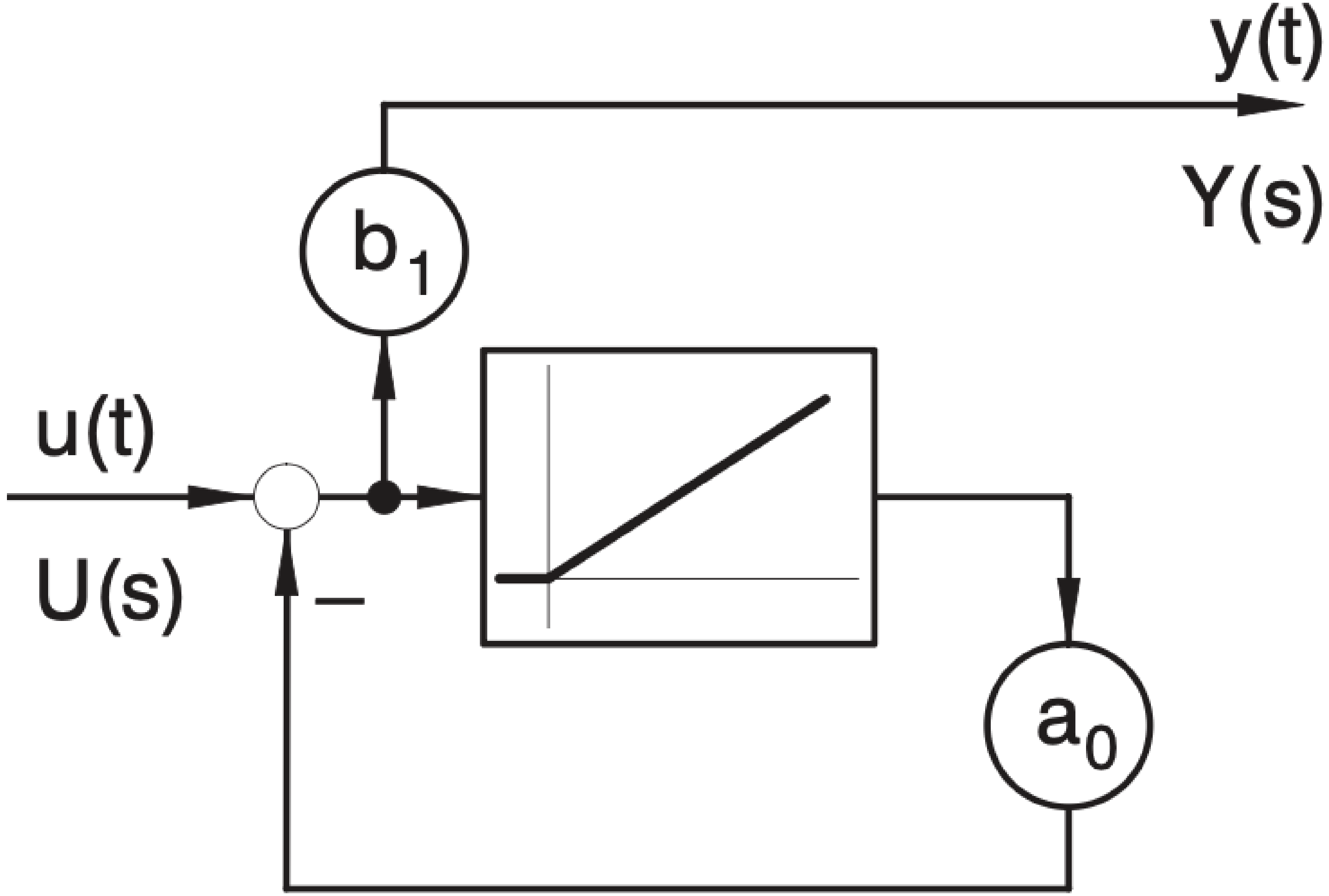


j Im{G}
ω
G
0
$\frac{T_2}{T_1}$
Re{G}



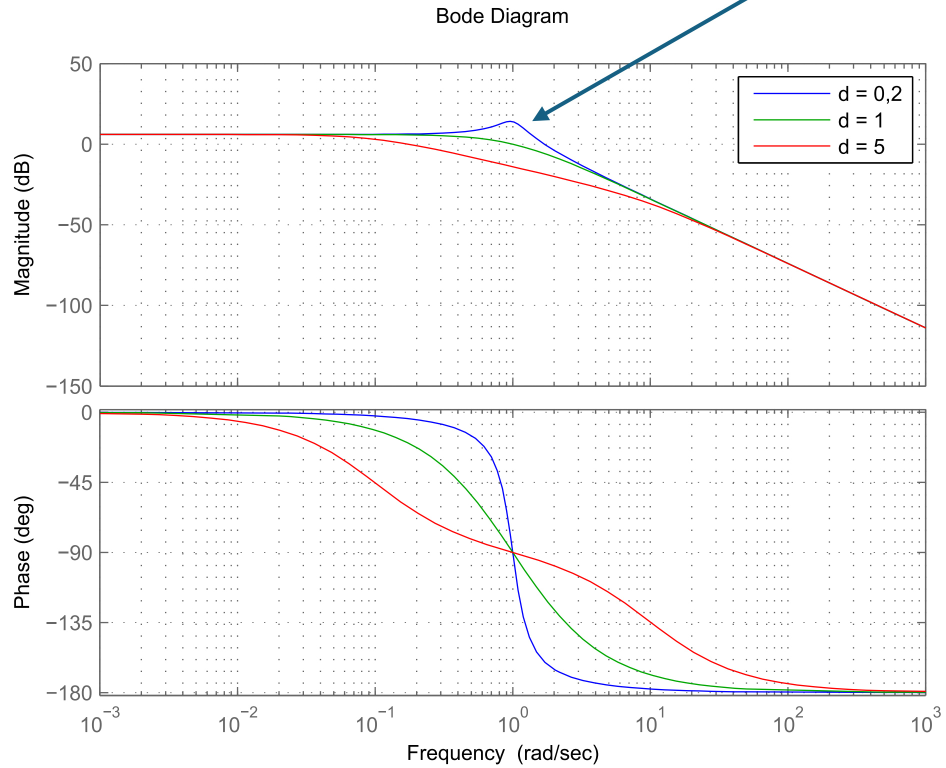
jω
s
σ
$-\frac{1}{T_1}$

DT1 Signal Flow Diagram



Second-Order Lag Element (PT2)

Resonance
(complex poles)



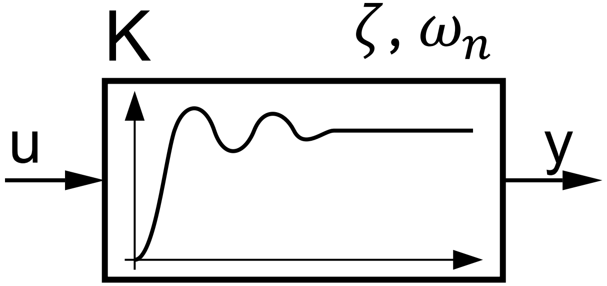
Consider $n = 2$, with $b_1 = b_2 = 0$:

$$\underbrace{\frac{a_2}{a_0}}_{\frac{1}{\omega_0^2}} \frac{d^2y(t)}{dt^2} + \underbrace{\frac{a_1}{a_0}}_{\frac{2\zeta}{\omega_n}} \frac{dy(t)}{dt} + y(t) = \underbrace{\frac{b_0}{a_0}}_{K} u(t)$$

$$Y(s) \cdot \left(\frac{a_2}{a_0} s^2 + \frac{a_1}{a_0} s + 1\right) = \frac{b_0}{a_0} \cdot U(s)$$

$$\boxed{G(s) = \frac{K}{\frac{1}{\omega_n^2} s^2 + \frac{2\zeta}{\omega_n} s + 1}}$$

Step response:



PT2 as Product of Two PT1 Elements

$G(s) = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}$ can be written as product of two PT1 elements:

$$G(s) = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

individual time constants

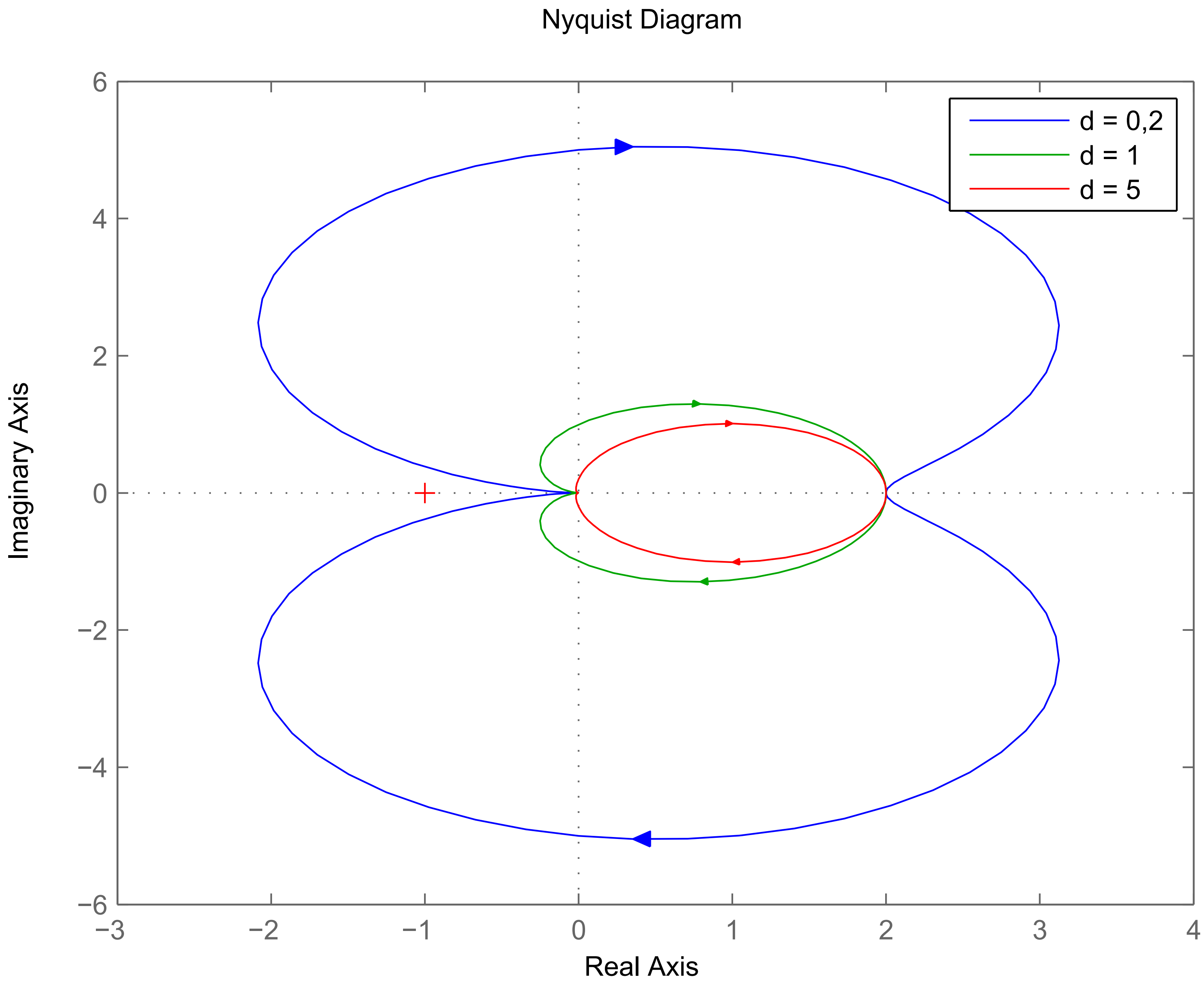
$$\Rightarrow \frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1 = \tau_1\tau_2 s^2 + (\tau_1 + \tau_2)s + 1$$

$$\Rightarrow \frac{1}{\omega_n^2} = \tau_1\tau_2, \quad \frac{2\zeta}{\omega_n} = (\tau_1 + \tau_2)$$

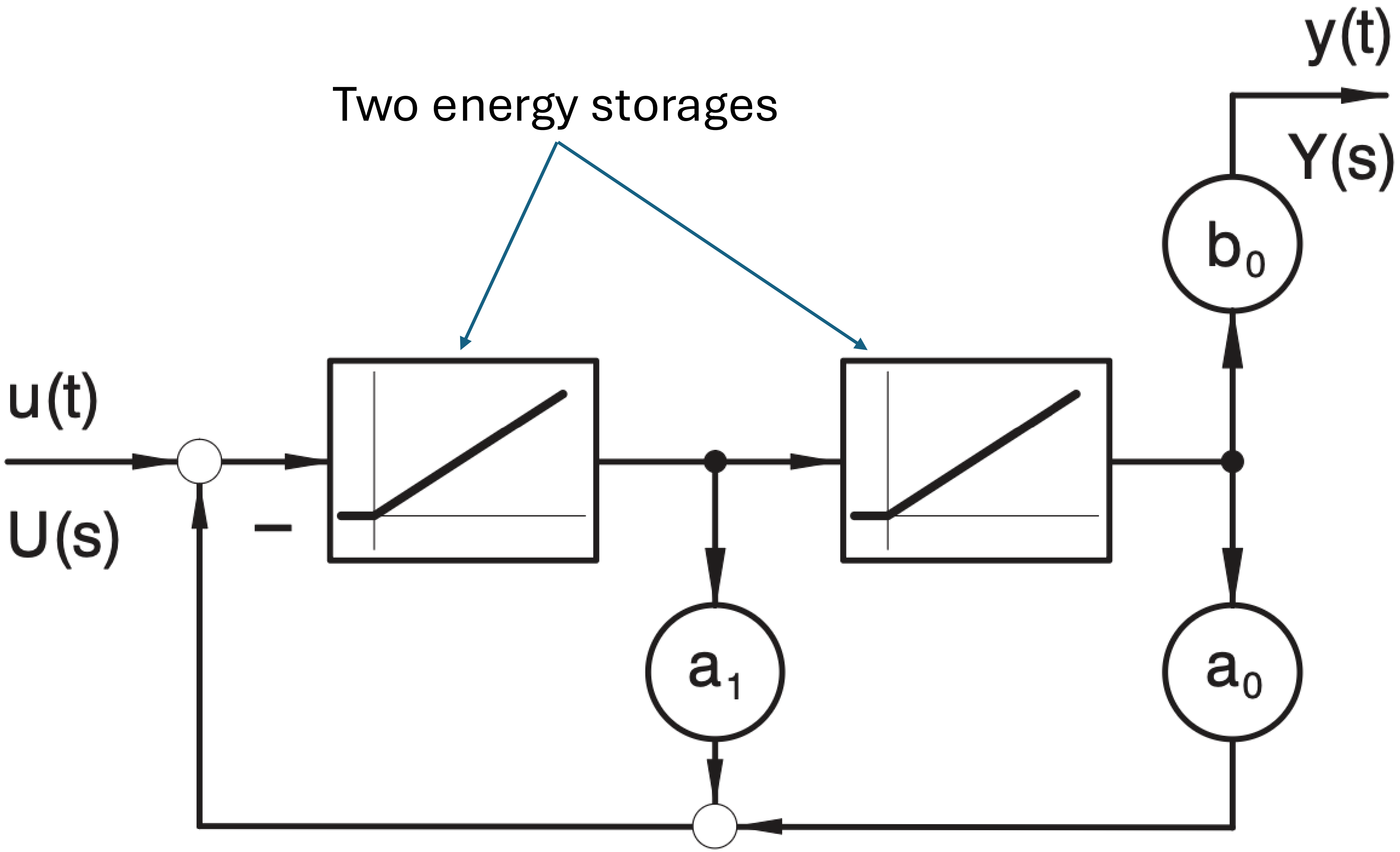
$$\Rightarrow \zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1\tau_2}}$$

for real PT1 elements: $\tau_1 + \tau_2 \geq 2\sqrt{\tau_1\tau_2}$ (arithmetic-geometric mean inequality)

→ overdamped ($\zeta < 1$) or critically damped ($\zeta = 1$) for $\underbrace{\tau_1 = \tau_2}_{\text{fastest non-oscillatory response}}$


Nyquist Diagram
d = 0,2
d = 1
d = 5
Imaginary Axis
Real Axis
6
4
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
4

PT2 Signal Flow Diagram



All-Pass Systems

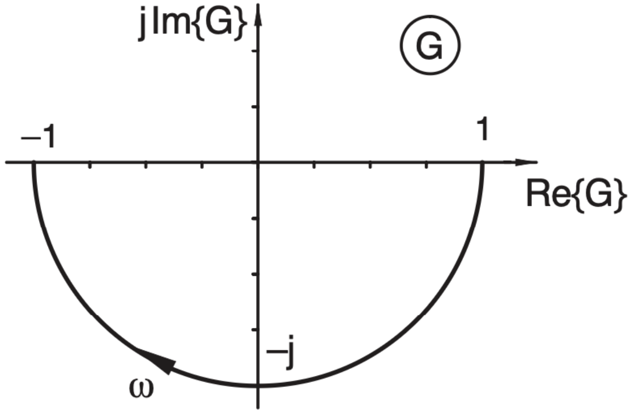
Magnitude constant for all frequencies ($|G(j\omega)| = 1$), but phase depends on frequency

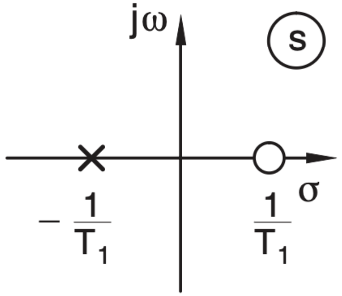
coefficients: $b_k = (-1)^{n-k} a_{n-k}$

→ Zeros are mirror images of the poles with respect to the imaginary axis.

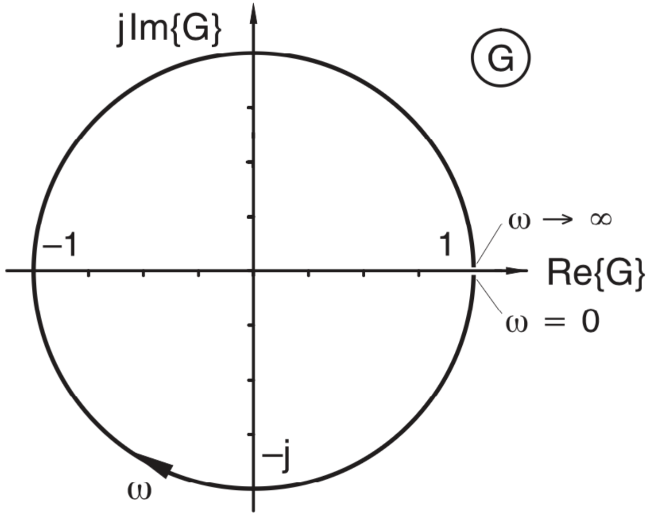
Important in signal processing, e.g., for runtime alignment

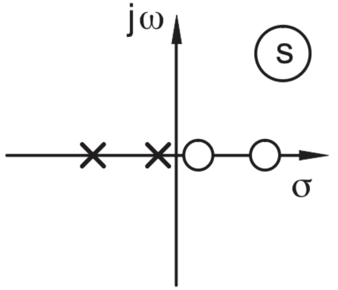
first-order all-pass: $G(s) = \frac{s-a}{s+a}$

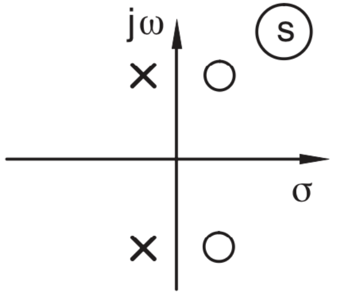




second-order all-pass: $G(s) = \frac{a_2 s^2 - a_1 s + a_0}{a_2 s^2 + a_1 s + a_0}$







Systems with Dead Time

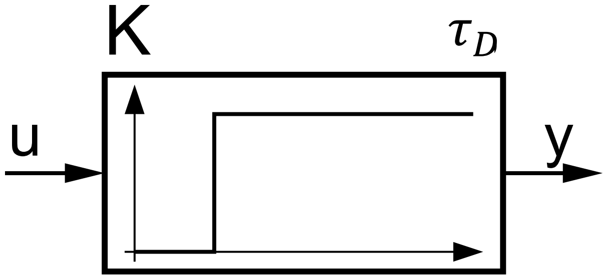
The transfer function of a system can be extended by multiplying it with the exponential term that represents a time delay:

$$G(s) = G_0(s) \cdot e^{-s \cdot \tau_D}$$

original transfer function

time delay (aka dead time)

Step response (P element with dead time):



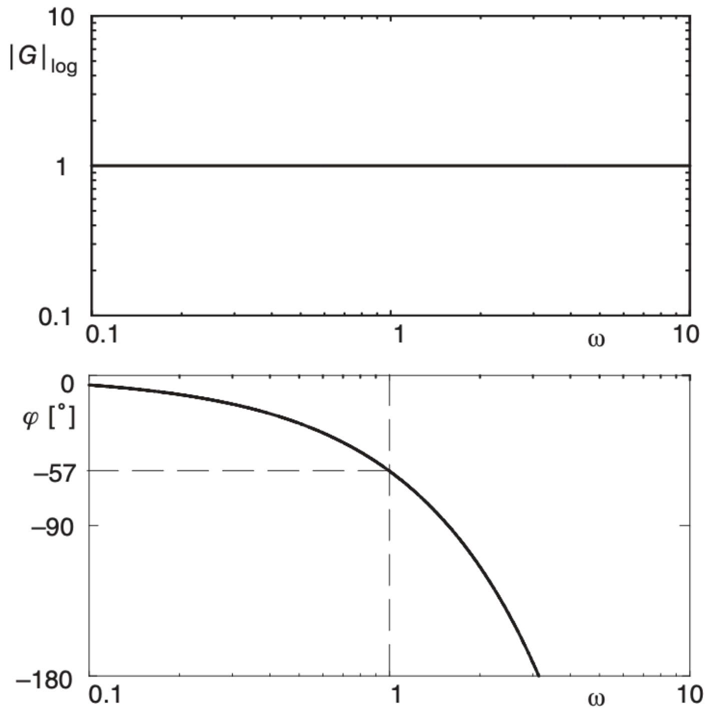
Example: PT1 with dead time

$$G(s) = \frac{K}{1+\tau \cdot s} \cdot e^{-s \cdot \tau_D}$$

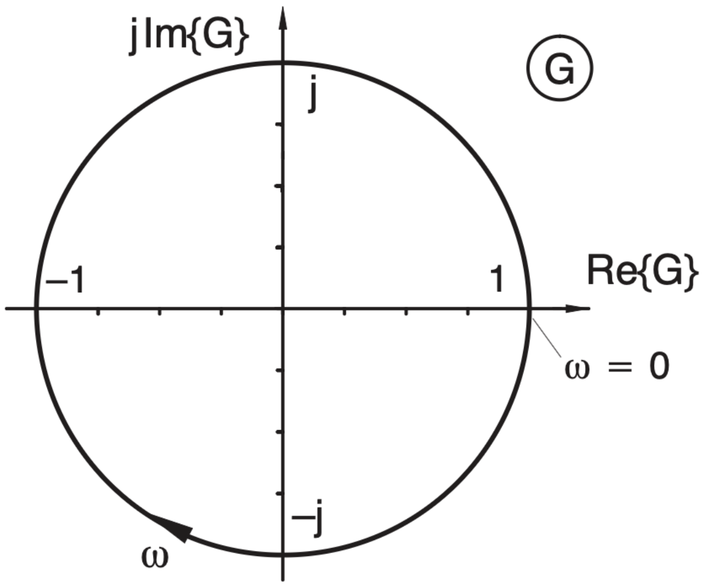
Mainly suitable for frequency response analysis (e.g., Bode plots)

Dead time is one of the most common reasons for instability and poor control behavior.

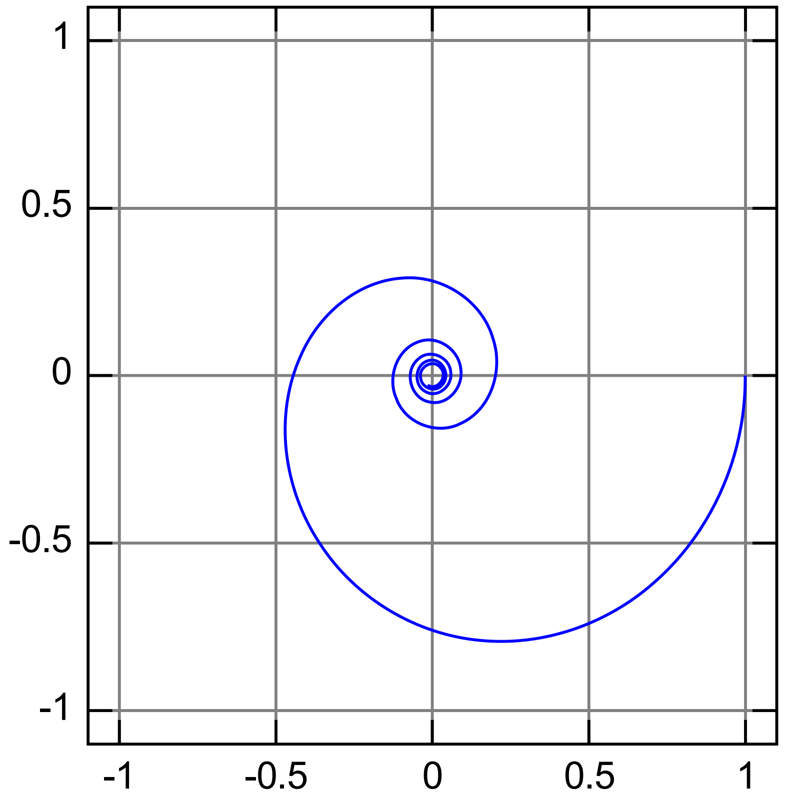
Bode diagram for system with dead time:



Nyquist diagram for system with dead time:



Nyquist diagram for PT1 with dead time:



Delay = additional phase shift that grows linearly with ω

→ Dead time hides in the amplitude, but reveals itself in the phase.

System Identification with Bode Plots

Experiment: frequency response measurement → Bode plots

- Magnitude plot: sum of dB contributions of all components
- Phase plot: sum of phase contributions

Decompose system into standard building blocks (integrator, PT1, gain, ...) → straight line segments

System identification procedure (construct transfer function):

1. Determine initial slope (low frequencies) → indicates integrators/differentiators
2. Identify corner frequencies → points where slope changes
3. Analyze slope changes: $-20\ \mathrm{dB}\ /\mathrm{decade}$ → pole, $+20\ \mathrm{dB}\ /\mathrm{decade}$ → zero
4. Determine time constants $\omega_c = 1/\tau$
5. Determine gain K from absolute magnitude level

slope = structure, corner frequency = dynamics, level = gain

Example

Check magnitude plot:

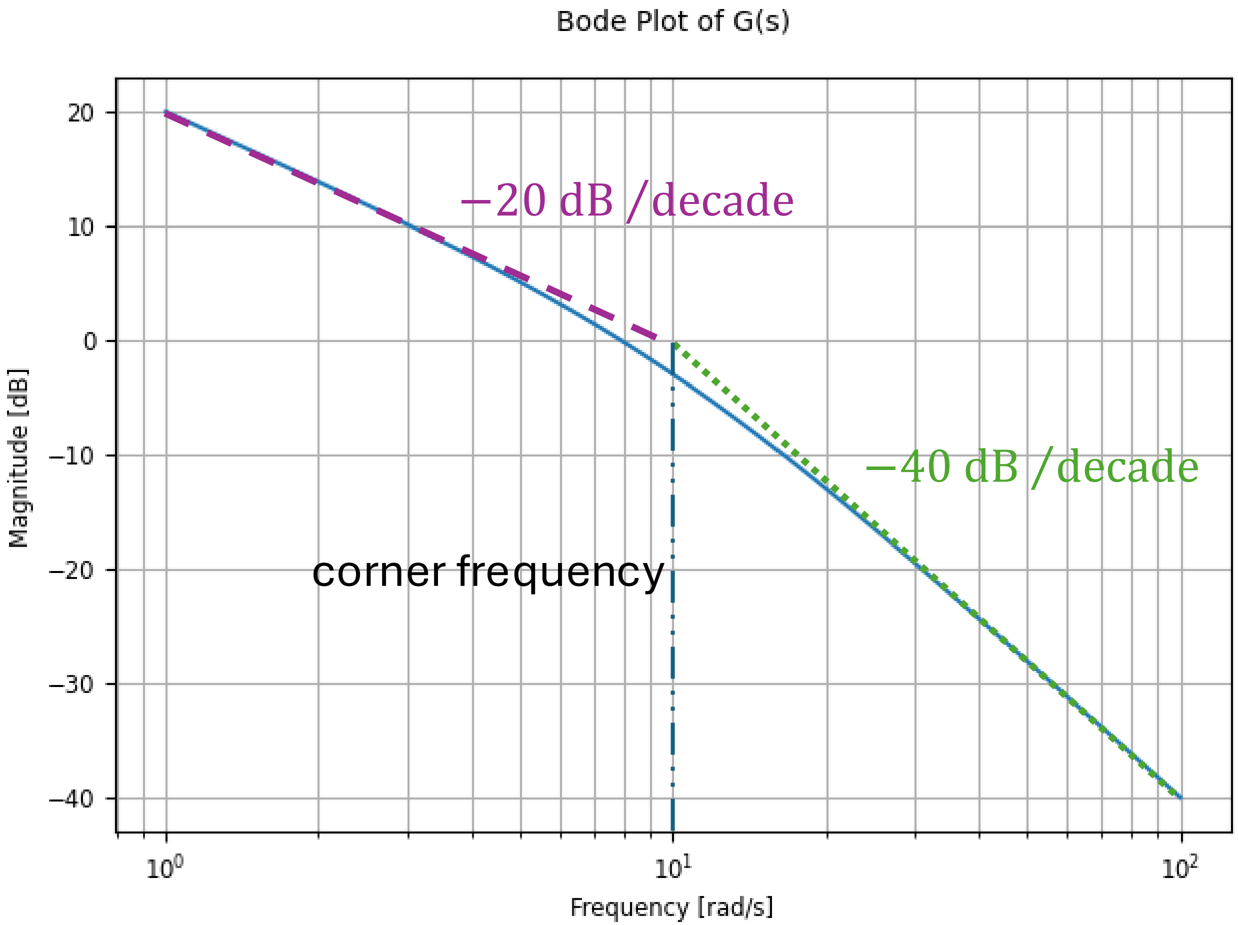
- For small frequencies: $-20\ \mathrm{dB}\ /\mathrm{decade}$ → one more pole than zeros (often integrator)
- For large frequencies: $-40\ \mathrm{dB}\ /\mathrm{decade}$ → two poles overall
- $20\ \mathrm{dB}$ at $\omega = 1$

→ Gain $K = 10$ $(20\ \mathrm{dB} = 20\log_{10}(K))$

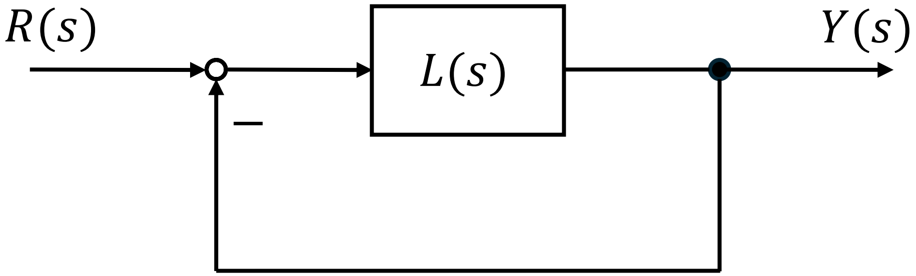
→ One pole always active → integrator $1/s$

→ One more pole comes later: kink with corner frequency at $\omega_c \approx 10\ \mathrm{rad}/s$ → PT1 $\frac{1}{1+0.1s}$

$$\Rightarrow G(s) = \frac{10}{s(1+0.1s)}$$



Control Loop

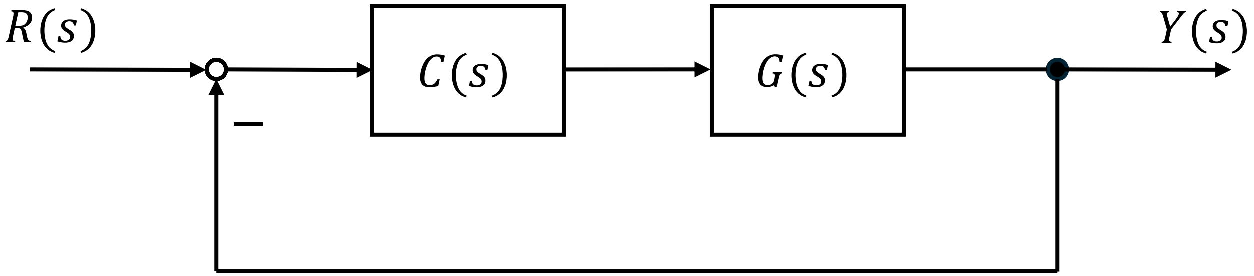


$$Y(s) = L(s) \cdot (R(s) - Y(s))$$

$$\Rightarrow Y(s) \cdot (1 + L(s)) = L(s) \cdot R(s)$$

$$\Rightarrow \frac{Y(s)}{R(s)} = \frac{L(s)}{1 + L(s)}$$

closed-loop transfer function
for negative feedback



$$\frac{Y(s)}{R(s)} = \frac{C(s) \cdot G(s)}{1 + C(s) \cdot G(s)}$$

$G(s)$ above combines controller $C(s)$ and plant $G(s)$

PID Controller Transfer Function

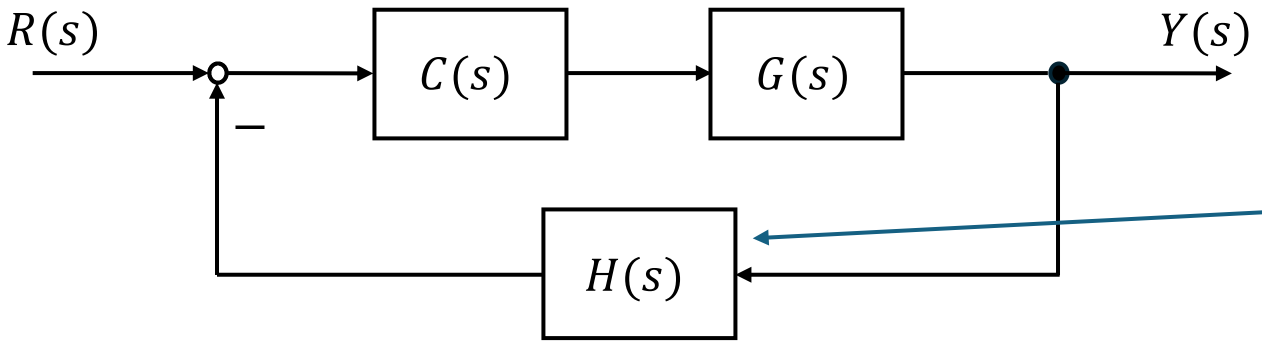
$$C(s) = K_p + \frac{K_i}{s} + K_d \cdot s$$

- K_p: increases loop gain → faster response, smaller steady-state error, may increase overshoot
- K_i: adds integrator → eliminates steady-state error, may reduce stability margin
- K_d: adds damping → reduces overshoot, improves transient behavior

Controller design:

- Shape the closed-loop poles to obtain the desired dynamics
- Achieved by selecting and tuning controller parameters (move the closed-loop poles)

Hidden Assumptions

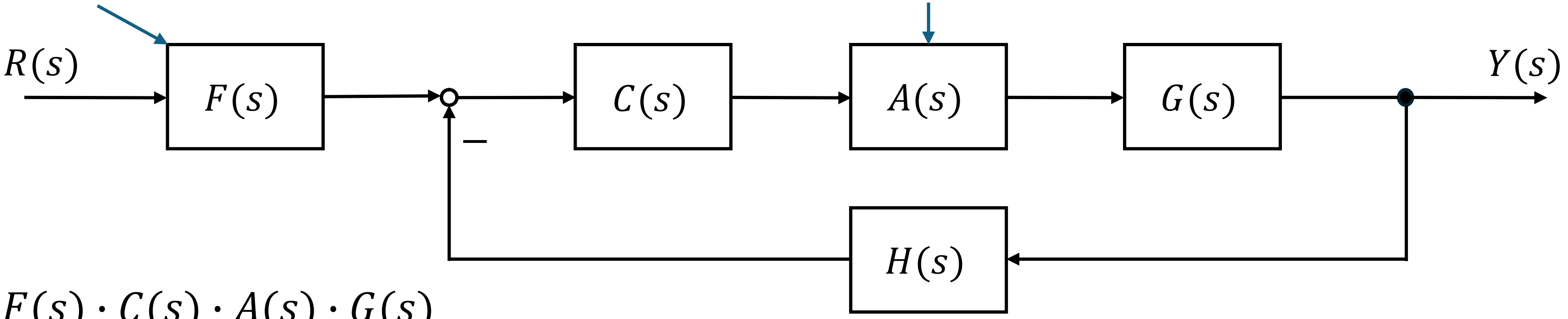


$$\frac{Y(s)}{R(s)} = \frac{C(s) \cdot G(s)}{1 + C(s) \cdot G(s) \cdot H(s)}$$

sensor dynamics: If sensor is fast, one can assume unity feedback.

reference prefilter: If reference shaping doesn't affect stability, one can treat it separately.

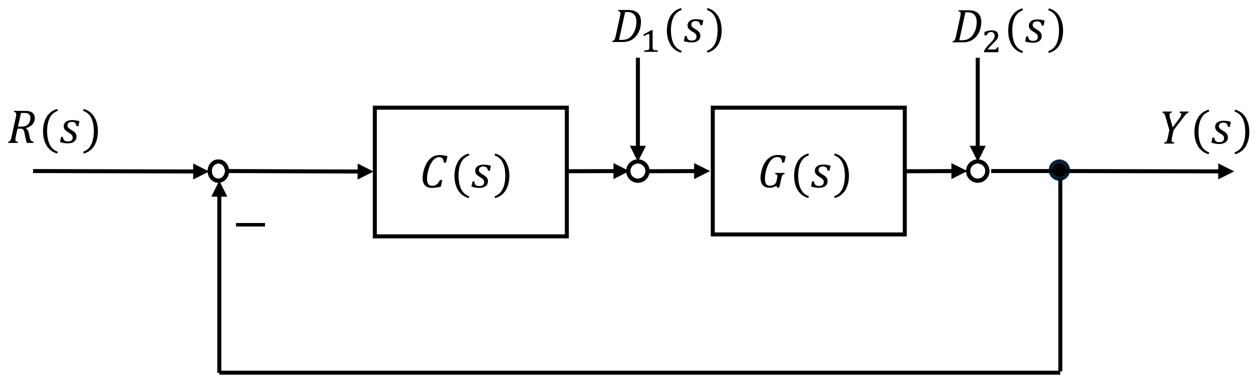
actuator dynamics: If actuator bandwidth ≫ closed-loop bandwidth, one can ignore it.



$$\frac{Y(s)}{R(s)} = \frac{F(s) \cdot C(s) \cdot A(s) \cdot G(s)}{1 + C(s) \cdot A(s) \cdot G(s) \cdot H(s)}$$

Always model what can affect poles

Disturbances



$$Y(s) = \frac{C(s) \cdot G(s)}{1 + C(s) \cdot G(s)} \cdot R(s) + \frac{G(s)}{1 + C(s) \cdot G(s)} \cdot D_1(s) + \frac{1}{1 + C(s) \cdot G(s)} \cdot D_2(s)$$

Output as superposition of input and disturbances

Exercise 1

Find the Laplace transform of the following signals:

- $f(t) = \sin(4t) + 5t$

- $f(t) = \frac{d\left(3e^{-2t}\right)}{dt}$

Exercise 2

Given the differential equation:

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = u(t)$$

- Determine the transfer function of this system.
- Calculate the poles and zeros of this transfer function.
- For a unit step input, calculate the solution in the time domain.

Exercise 3

Given the differential equation:

$$\dddot{y}(t) + 2\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 2\dot{u}(t) + u(t)$$

- Determine the transfer function of this system.
- Calculate the poles and zeros of this transfer function.

Exercise 4

Find the steady-state value of the unit step response for:

$$G(s) = \frac{5}{s+5}$$

Exercise 5

A system consists of:

- Forward path: $G(s) = \frac{2}{s+1}$
- Feedback: $H(s) = 1$

Find the closed-loop transfer function.

Exercise 6

Given the transfer function:

$$G(s) = \frac{1}{s+1}$$

Find the magnitude of the frequency response at $\omega = 1$.

Exercise 7

Consider the Bode plots for the system:

$$G(s) = \frac{10}{s}$$

- What is the slope of the magnitude plot?
- What is the phase?

Exercise 8

Given a first-order system with the transfer function:

$$G(s) = \frac{1}{1+0.1s}$$

Estimate the bandwidth.

Exercise 9

Consider a plant with transfer function:

$$G(s) = \frac{1}{s(s+1)}$$

A PID controller in parallel form is used with parameters:

$$K_p = 3, K_i = 2, K_d = 1$$

Assume unity feedback.

- Determine the open-loop transfer function.
- Derive the closed-loop transfer function.